

We can use AWS Clean Rooms to share data with partners or vice-versa as plain text, hashed data dump on S3 bucket, or pre-encrypted data. It can be configured easily for multiple collaborators without revealing raw data of the partner organisation. We can perform joint analysis with AWS Clean Rooms for research or market analysis, or improved reporting.

This kind of service helps to play around with real customer data for data analytics in a sandbox setup, before the actual live data is experienced in public for data analysis. This helps to improve reporting and monitoring, as well as product development.

We can use templated queries for controlling data operations, flexible rules to handle data and its access management, and associate existing data (e.g., historical data analytics) for collaborative data analytics. We can use AWS machine learning services like SageMaker and Quicksight for deep dive analytics and insights from multiple collaborative data using AWS Clean Rooms.

Data shared by partners and clients in AWS Clean Rooms is fully secure and protected from unauthorised access. It is encrypted, safe to store,

and safe to clean and remove from AWS Clean Rooms. It supports file formats like csv and parquet, and can be pre-encrypted before being transported for data analytics.

Since it has been released only recently, it is available only in selected regions; we can expect general availability (GA) of AWS Clean Rooms soon.

Visual data exploration in Amazon Neptune

Graph databases are popular for both transactional data (OLTP) and analytical data (OLAP) visualisation, for a quick analysis of entity relations and data dependencies. Amazon Neptune (Figure 3) is a popular, fully-managed graph database that is fast and reliable for interpreting data visually. It helps to visually interpret highly connected data sets, prepare knowledge graphs and navigate data easily.

Amazon Neptune is a low-latency graph database with multi-regional and multi-availability zone replication, continuous backup to Amazon S3 bucket for file streaming and data feeds, and point-in-time recovery and read replicas for high throughput with queries. It uses the Apache TinkerPop

framework for graph visualisation of data set and graph analytic capabilities, and uses Apache TinkerPop Gremlin as a graph traversal language to express complex traversals visually.

Amazon Neptune supports querying through W3C's SPARQL and Neo4j openCypher to navigate through connected data sets with the support of Apache TinkerPop Gremlin. It also supports Open Graph API with Gremlin and SPARQL for fully-managed graphical data sets.

Recently, Amazon Neptune announced native integration and support for open source Graph Explorer for visually interpreting data sets through a web application interface based on ReactJS. It can be deployed as a Dockerized image in Amazon EC2 or Amazon ECS, and supports SPARQL and Apache TinkerPop Gremlin.

We can prepare data sets in Amazon Neptune through queries and use 'send to explorer' to connect to the Data Explorer source from Graph Explorer, in order to visually interpret the data sets and navigate the collection easily without querying data every time.

Graph Explorer reduces the work involved in querying and data set preparation, and is useful for non-

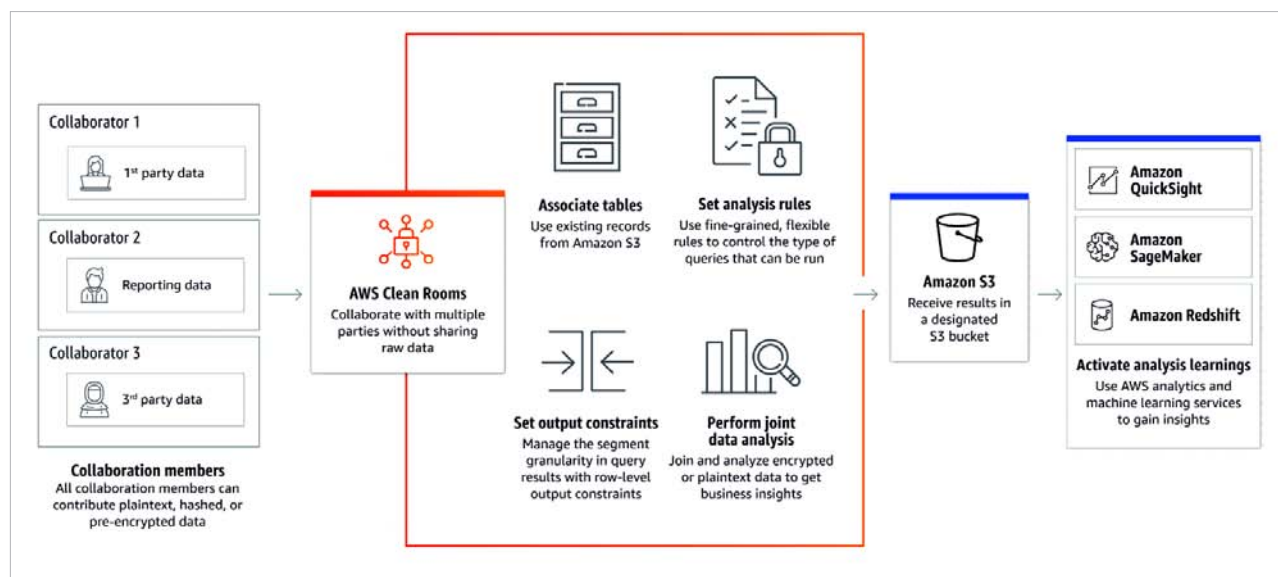


Figure 2: Amazon Clean Rooms